Comparison: `downup_ac` vs `downup_ac_pop` (rural sample)

June 23, 2026

This document compares all rural results under the original treatment `downup_ac` and the population-weighted variant `downup_ac_pop`, across the full rural-replication, stacked event-study, and dCDH pipelines.

† **HonestDiD panels (Sections 2–6):** each treatment row shows a 2×2 grid. Top row: original event-study coefficients (`_ori`) and their rotation-based version (`_rotated`). Bottom row: *relative-magnitudes* sensitivity bounds from `HonestDiD::createSensitivityResults_relativeMagnitudes` applied to the **original** coefficients (`_honest1`) and to the **rotated** coefficients (`_rot_honest1`). Other files on disk but not shown: `_honest2` and `_rot_honest2` (smoothness-restriction bound on original and rotated, respectively).

Missing figures or tables are flagged inline as *(pending: . . .)* and do not break compilation.

Contents

1 DiD Regression Tables

Each subsection shows the original `downup_ac` table (red rule) followed by the `downup_ac_pop` table (green rule). Tables are `\input`'d from the Dropbox `tables/` folder; the stems follow the pattern `<name>_rural.tex` / `<name>_rural_acpop.tex`.

1.1 Main DiD (`_main_1`)

Original (`downup_ac`)

	(1)	(2)	(3)	(4)
	Number of Fires (in 1,000 units)			
Down > Up	-12.370***	-12.323***	-8.908***	-7.421***
	(2.044)	(1.676)	(1.396)	(1.424)
Observations	9,216,000	9,216,000	9,214,800	9,214,800
N Assembly Constituencies	808	808	808	808
Month-Year FE	N	Y	N	N
AC FE	N	Y	N	N
AC × Month-Year FE	N	N	Y	Y
Grid FE	N	N	N	Y
Mean DV	179.948	179.948	179.948	179.948

New (`downup_ac_pop`)

	(1)	(2)	(3)	(4)
	Number of Fires (in 1,000 units)			
Down > Up Population	-6.418***	-10.660***	-8.865***	-7.165***
	(1.996)	(1.747)	(1.387)	(1.360)
Observations	9,208,200	9,208,200	9,207,000	9,207,000
N Assembly Constituencies	808	808	808	808
Month-Year FE	N	Y	N	N
AC FE	N	Y	N	N
AC × Month-Year FE	N	N	Y	Y
Grid FE	N	N	N	Y
Mean DV	170.747	170.747	170.747	170.747

1.2 Bureaucrat × Politician DiD (_main_3)

Original (downup_ac)

	(1)	(2)	(3)	(4)	(5)
	Number of Fires (in 1,000 units)				
Down> Up Bureaucrat	-29.445***	-25.665***	-4.396*	2.504	-19.201***
	(2.869)	(3.605)	(2.580)	(2.610)	(3.350)
Down> Up Politician	-10.238***	-16.880***	-15.303***	-8.073***	-5.572**
	(3.033)	(2.733)	(2.140)	(1.992)	(2.182)
Down> Up Pol. × Down> Up Bur.	10.884***	18.840***	11.708***	0.015	7.070**
	(3.963)	(3.486)	(2.739)	(2.422)	(2.794)
Observations	9,216,000	9,216,000	9,214,800	9,214,800	9,216,000
N Assembly Constituencies	808	808	808	808	808
N Districts	160	160	160	160	160
Month-Year FE	N	Y	N	N	N
AC FE	N	Y	N	N	N
AC × Month-Year FE	N	N	Y	Y	N
District × Month-Year FE	N	N	N	N	Y
Grid FE	N	N	N	Y	Y
Mean DV	193.722	193.722	193.722	193.722	193.722

New (downup_ac_pop)

	(1)	(2)	(3)	(4)	(5)
	Number of Fires (in 1,000 units)				
Down> Up Bureaucrat	-28.266***	-27.293***	-4.821*	1.765	-18.812***
	(2.821)	(3.517)	(2.587)	(2.640)	(3.287)
Down> Up Politician (Pop)	-0.144	-15.607***	-14.967***	-7.946***	-4.894**
	(2.975)	(2.887)	(2.131)	(1.916)	(2.114)
Down> Up Pol. (Pop) × Down> Up Bur.	4.176	20.198***	11.493***	0.585	6.159**
	(3.887)	(3.593)	(2.693)	(2.323)	(2.760)
Observations	9,216,000	9,216,000	9,214,800	9,214,800	9,216,000
N Assembly Constituencies	808	808	808	808	808
N Districts	160	160	160	160	160
Month-Year FE	N	Y	N	N	N
AC FE	N	Y	N	N	N
AC × Month-Year FE	N	N	Y	Y	N
District × Month-Year FE	N	N	N	N	Y
Grid FE	N	N	N	Y	Y
Mean DV	184.328	184.328	184.328	184.328	184.328

1.3 Protest 5 km DiD (_main_4)

Original (downup_ac)

	(1)	(2)	(3)	(4)	(5)	(6)
	Number of Fires (in 1,000 units)					
Post × Protest	19.171**	76.917***	55.240***	23.512**	82.086***	60.585***
	(8.525)	(11.295)	(11.027)	(11.565)	(13.855)	(13.624)
Down > Up				-11.977***	-12.010***	-12.021***
				(4.287)	(4.259)	(4.260)
Post × Down > Up				6.114**	6.135**	6.257**
				(2.648)	(2.669)	(2.648)
Protest × Down > Up				22.614	22.599	22.581
				(27.573)	(27.353)	(27.296)
Post × Protest × Down > Up				-8.595	-10.209	-10.567
				(14.332)	(14.333)	(14.292)
Observations	148,171,030	148,171,030	148,171,030	148,171,030	148,171,030	148,171,030
N Assembly Constituencies	808	808	808	808	808	808
Grid FE	Y	Y	Y	Y	Y	Y
Relative Time FE	Y	Y	Y	Y	Y	Y
Legislature FE	N	Y	Y	N	Y	Y
Province Trend FE	N	N	Y	N	N	Y
Mean DV	439.477	439.477	439.477	439.477	439.477	439.477
Mean DV2 (Down>Up=1)				437.78	437.78	437.78

New (downup_ac_pop)

	(1)	(2)	(3)	(4)	(5)	(6)
	Number of Fires (in 1,000 units)					
Post × Protest	19.135**	76.951***	55.291***	23.284**	82.315***	60.622***
	(8.534)	(11.308)	(11.041)	(11.424)	(13.731)	(13.454)
Down > Up (Pop)				-8.193**	-8.238**	-8.247**
				(4.018)	(3.985)	(3.986)
Post × Down > Up (Pop)				10.019***	9.746***	9.824***
				(2.938)	(2.909)	(2.909)
Protest × Down > Up (Pop)				36.506	36.478	36.307
				(28.512)	(28.291)	(28.240)
Post × Protest × Down > Up (Pop)				-8.101	-10.485	-10.440
				(14.794)	(14.847)	(14.854)
Observations	148,049,467	148,049,467	148,049,467	148,049,467	148,049,467	148,049,467
N Assembly Constituencies	808	808	808	808	808	808
Grid FE	Y	Y	Y	Y	Y	Y
Relative Time FE	Y	Y	Y	Y	Y	Y
Legislature FE	N	Y	Y	N	Y	Y
Province Trend FE	N	N	Y	N	N	Y
Mean DV	439.77	439.77	439.77	439.77	439.77	439.77
Mean DV2 (Down>Up=1)				447.438	447.438	447.438

1.4 Politician Characteristics DiD (_main_5)

Original (downup_ac)

	(1)	(2)	(3)	(4)	(5)	(6)
	Number of Fires (in 1,000 units)					
Post × Agriculturalist	29.244***	6.288**	7.316***	30.298***	7.145**	8.084**
	(3.659)	(2.854)	(2.827)	(3.958)	(3.251)	(3.260)
Down > Up				-14.505***	-14.427***	-19.519***
				(2.276)	(2.258)	(2.376)
Post × Down > Up				2.545	2.424	2.068
				(2.196)	(2.190)	(2.267)
Agriculturalist × Down > Up				9.813***	9.548***	12.249***
				(3.492)	(3.479)	(3.592)
Post × Agriculturalist				-2.089	-1.696	-1.514
				(3.277)	(3.267)	(3.343)
Observations	21,882,509	21,882,509	21,882,509	21,882,509	21,882,509	21,882,509
N Assembly Constituencies	658	658	658	658	658	658
Grid FE	Y	Y	Y	Y	Y	Y
Relative Time FE	Y	Y	Y	Y	Y	Y
Legislature FE	N	Y	Y	N	Y	Y
Province Linear Time Trend FE	N	N	Y	N	N	Y
Mean DV	119.847	119.847	119.847	119.847	119.847	119.847
Mean DV (Down > Up=1)	119.847	119.847	119.847	119.36	119.36	119.36

New (downup_ac_pop)

	(1)	(2)	(3)	(4)	(5)	(6)
	Number of Fires (in 1,000 units)					
Post × Agriculturalist	29.306***	6.326**	7.362***	28.090***	5.332	6.224*
	(3.662)	(2.859)	(2.832)	(4.085)	(3.401)	(3.459)
Down > Up (Pop)				-8.738***	-9.220***	-13.539***
				(2.898)	(2.869)	(3.092)
Post × Down > Up (Pop)				-0.670	0.158	-0.122
				(2.688)	(2.659)	(2.813)
Agriculturalist × Down > Up (Pop)				10.259**	10.421**	12.278***
				(4.266)	(4.210)	(4.473)
Post × Agriculturalist				2.409	1.947	2.239
				(3.993)	(3.941)	(4.116)
Observations	21,854,616	21,854,616	21,854,616	21,854,616	21,854,616	21,854,616
N Assembly Constituencies	650	650	650	650	650	650
Grid FE	Y	Y	Y	Y	Y	Y
Relative Time FE	Y	Y	Y	Y	Y	Y
Legislature FE	N	Y	Y	N	Y	Y
Province Linear Time Trend FE	N	N	Y	N	N	Y
Mean DV	119.919	119.919	119.919	119.919	119.919	119.919
Mean DV (Down > Up=1)	119.919	119.919	119.919	122.648	122.648	122.648

1.5 Alternative Dependent Variables (_app_7)

Original (`downup_ac`)

	(1) Any Fire	(2) Log (N) Fires	(3) Mean Brightness
Down > Up	-0.0022***	-0.0029***	-0.7228***
	(0.0003)	(0.0004)	(0.0898)
Observations	9,214,800	9,214,800	9,214,800
N Assembly Constituencies	808	808	808
Grid FE	Y	Y	Y
Assembly × Month-Year FE	Y	Y	Y
Mean DV	0.056	0.068	18.67

New (`downup_ac_pop`)

	(1) Any Fire	(2) Log (N) Fires	(3) Mean Brightness
Down > Up (Pop)	-0.0020***	-0.0027***	-0.6677***
	(0.0003)	(0.0004)	(0.0859)
Observations	9,207,000	9,207,000	9,207,000
N Assembly Constituencies	808	808	808
Grid FE	Y	Y	Y
Assembly × Month-Year FE	Y	Y	Y
Mean DV	0.054	0.065	17.92

1.6 DiD by Year (_app_8)

Original (downup_ac)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Number of Fires (in 1,000 units)									
	2012/2013	2013/2014	2014/2015	2015/2016	2016/2017	2017/2018	2018/2019	2019/2020	2020/2021	2021/2022
Down>Up	-6.644*	-10.890***	-6.997*	-6.550	-10.716**	-9.904***	-10.443***	-5.322**	-5.475	-1.201
	(3.864)	(3.708)	(4.199)	(4.051)	(4.575)	(3.816)	(3.676)	(2.578)	(3.627)	(3.979)
Observations	921,480	921,480	921,480	921,480	921,480	921,480	921,480	921,480	921,480	921,480
N Assembly Constituencies	808	808	808	808	808	808	808	808	808	808
Grid FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Assembly × Month-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Mean DV	178.96	168.38	174.28	175.15	210.74	176	190.14	126.32	193.31	206.18

New (downup_ac_pop)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Number of Fires (in 1,000 units)									
	2012/2013	2013/2014	2014/2015	2015/2016	2016/2017	2017/2018	2018/2019	2019/2020	2020/2021	2021/2022
Down>Up (Pop)	-6.795*	-9.374***	-5.759	-6.111	-11.088**	-9.917***	-9.445***	-5.438**	-6.374*	-2.411
	(3.733)	(3.547)	(4.027)	(3.844)	(4.405)	(3.709)	(3.487)	(2.589)	(3.479)	(3.892)
Observations	920,700	920,700	920,700	920,700	920,700	920,700	920,700	920,700	920,700	920,700
N Assembly Constituencies	808	808	808	808	808	808	808	808	808	808
Grid FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Assembly × Month-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Mean DV	170.99	160.23	165.25	167.04	199.68	166.34	181.34	119	182.8	194.79

1.7 DiD by State (_app_9)

Original (downup_ac)

	(1)	(2)	(3)	(4)
	Number of Fires (in 1,000 units)			
	Bihar	Haryana	Punjab	Uttar Pradesh
Down>Up	1.167	-18.434***	-19.560***	-3.061***
	(1.688)	(5.862)	(5.505)	(0.903)
Observations	1,634,640	1,040,400	1,227,840	5,311,920
N Assembly Constituencies	237	84	107	380
Grid FE	Y	Y	Y	Y
Assembly × Month-Year FE	Y	Y	Y	Y
Mean DV	43.423	181.541	780.009	47.893

New (downup_ac_pop)

	(1)	(2)	(3)	(4)
	Number of Fires (in 1,000 units)			
	Bihar	Haryana	Punjab	Uttar Pradesh
Down>Up (Pop)	0.634	-15.775***	-17.792***	-3.807***
	(1.479)	(5.480)	(5.438)	(0.893)
Observations	1,633,200	1,040,040	1,226,760	5,307,000
N Assembly Constituencies	237	84	107	380
Grid FE	Y	Y	Y	Y
Assembly × Month-Year FE	Y	Y	Y	Y
Mean DV	42.429	172.616	758.897	46.955

1.8 Rice Moderators DiD (_app_10)

Original (downup_ac)

	Number of Fires (in 1,000 units) - Rural Grids		
	(1)	(2)	(3)
Down>up AC	-9.176***	-6.234***	-6.705***
	(1.687)	(1.355)	(1.330)
Down>up AC × Above Median Rice Area	4.390		
	(3.053)		
Down>up AC × Above Median Harvested Rice Area		-2.659	
		(3.059)	
Down>up AC × Above Median Rice Production			-1.437
			(2.877)
Observations	9,214,800	9,214,800	9,214,800
N Assembly Constituencies	808	808	808
Grid FE	Y	Y	Y
AC × Month-Year FE	Y	Y	Y
Mean DV	179.948	179.948	179.948
Mean DV2	134.832	58.647	45.136
Mean DV3	244.426	332.632	318.29

New (downup_ac_pop)

	Number of Fires (in 1,000 units) - Rural Grids		
	(1)	(2)	(3)
Down>up AC (Pop)	-9.061***	-6.126***	-6.470***
	(1.591)	(1.256)	(1.232)
Down>up AC (Pop) × Above Median Rice Area	4.686		
	(2.919)		
Down>up AC (Pop) × Above Median Harvested Rice Area		-2.318	
		(2.925)	
Down>up AC (Pop) × Above Median Rice Production			-1.390
			(2.746)
Observations	9,207,000	9,207,000	9,207,000
N Assembly Constituencies	808	808	808
Grid FE	Y	Y	Y
AC × Month-Year FE	Y	Y	Y
Mean DV	170.747	170.747	170.747
Mean DV2	126.233	56.022	43.48
Mean DV3	233.813	317.672	305.768

1.9 Politician Char. Rice Moderators (_app_14)

Original (downup_ac)

	(1)	(2)	(3)	(4)
	Number of Fires (in 1,000 units)			
Post × Agriculturalist	7.316***	-1.750	-3.215	-1.090
	(2.827)	(3.181)	(2.608)	(2.669)
Post × Agriculturalist × Rice Areas		23.057***		
		(6.383)		
Post × Agriculturalist × Harvested Rice Area			27.063***	
			(6.481)	
Post × Agriculturalist × Rice Production				19.608***
				(6.012)
Observations	21,882,509	21,882,509	21,882,509	21,882,509
N Assembly Constituencies	658	658	658	658
Grid × Cohort FE	Y	Y	Y	Y
Relative Time FE	Y	Y	Y	Y
Legislature × Cohort FE	Y	Y	Y	Y
Province × Cohort Trend FE	Y	Y	Y	Y
Mean DV	119.847	119.847	119.847	119.847
Mean DV2		154.312	214.354	219.992
Mean DV3	119.847	97.361	63.277	48.334

New (downup_ac_pop)

	(1)	(2)	(3)	(4)
	Number of Fires (in 1,000 units)			
Post × Agriculturalist	7.362***	-1.706	-3.184	-1.058
	(2.832)	(3.186)	(2.614)	(2.675)
Post × Agriculturalist × Rice Areas		23.054***		
		(6.389)		
Post × Agriculturalist × Harvested Rice Area			27.089***	
			(6.487)	
Post × Agriculturalist × Rice Production				19.622***
				(6.018)
Observations	21,854,616	21,854,616	21,854,616	21,854,616
N Assembly Constituencies	650	650	650	650
Grid × Cohort FE	Y	Y	Y	Y
Relative Time FE	Y	Y	Y	Y
Legislature × Cohort FE	Y	Y	Y	Y
Province × Cohort Trend FE	Y	Y	Y	Y
Mean DV	119.919	119.919	119.919	119.919
Mean DV2		154.276	214.389	220.035
Mean DV3	119.919	97.501	63.35	48.379

1.10 Politician Char. DiD (_app_15)

Original (downup_ac)

	(1)	(2)	(3)
	Number of Fires (in 1,000 units)		
Post × Agriculturalist	29.244***	6.288**	7.316***
	(3.659)	(2.854)	(2.827)
Observations	21,882,509	21,882,509	21,882,509
N Assembly Constituencies	658	658	658
Grid × Cohort FE	Y	Y	Y
Relative Time FE	Y	Y	Y
Legislature × Cohort FE	N	Y	Y
Province × Cohort Trend FE	N	N	Y
Mean DV	119.847	119.847	119.847

New (downup_ac_pop)

	(1)	(2)	(3)
	Number of Fires (in 1,000 units)		
Post × Agriculturalist	29.306***	6.326**	7.362***
	(3.662)	(2.859)	(2.832)
Observations	21,854,616	21,854,616	21,854,616
N Assembly Constituencies	650	650	650
Grid × Cohort FE	Y	Y	Y
Relative Time FE	Y	Y	Y
Legislature × Cohort FE	N	Y	Y
Province × Cohort Trend FE	N	N	Y
Mean DV	119.919	119.919	119.919

1.11 Downwind Heatmap DiD (_app_20)

Original (downup_ac)

	(1)	(2)	(3)	(4)
	Number of Fires (in 1,000 units)			
Down > Up	-9.483**	-27.012***	-27.468***	-57.576***
	(4.045)	(6.143)	(5.978)	(20.941)
Down > Up × Rice Production	-93.847***	-18.482	-20.676	194.325***
	(20.740)	(13.998)	(13.863)	(38.473)
Observations	1,536,000	1,536,000	1,535,800	1,535,800
N Assembly Constituencies	808	808	808	808
Month-Year FE	N	Y	N	N
AC FE	N	Y	N	N
AC × Month-Year FE	N	N	Y	Y
Grid FE	N	N	N	Y
Mean DV	704.744	704.744	704.744	704.744
ymean2_clean	1444.898	1444.898	1444.898	1444.898

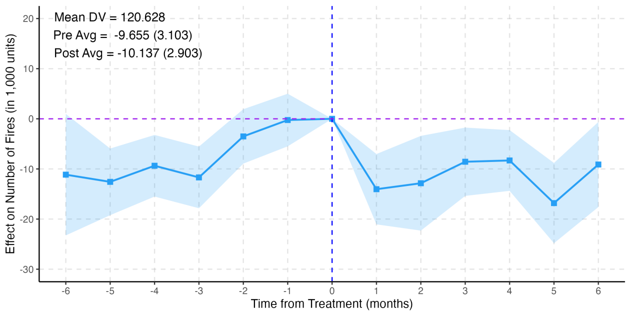
New (downup_ac_pop)

	(1)	(2)	(3)	(4)
	Number of Fires (in 1,000 units)			
Down > Up (Pop)	1.048	-22.553***	-26.455***	-41.817***
	(3.990)	(5.822)	(5.708)	(14.703)
Down > Up (Pop) × Rice Production	-86.479***	-21.657	-20.201	144.557***
	(20.376)	(14.073)	(13.867)	(26.704)
Observations	1,536,000	1,536,000	1,535,800	1,535,800
N Assembly Constituencies	808	808	808	808
Month-Year FE	N	Y	N	N
AC FE	N	Y	N	N
AC × Month-Year FE	N	N	Y	Y
Grid FE	N	N	N	Y
Mean DV	648.769	648.769	648.769	648.769
ymean2_clean	1284.496	1284.496	1284.496	1284.496

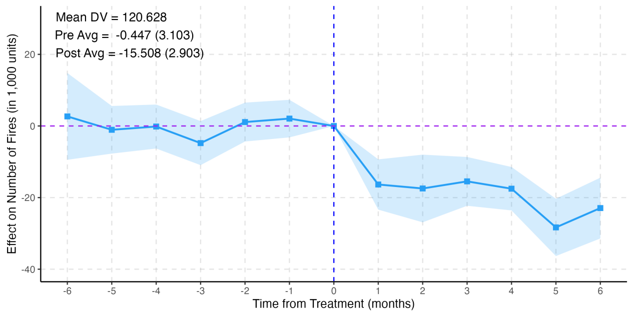
2 Main Event Study (_main_2)

Source: `_replication_rural/plotting_event_studies.R` and `_replication_rural_acpop/plotting_event_stud` Figures land in `tex/paper/figures/`. Each block: row 1 = `downup_ac`, row 2 = `downup_ac_pop`; each row is a 2×2 grid (original | rotated / HonestDiD-ori† | HonestDiD-rotated†).

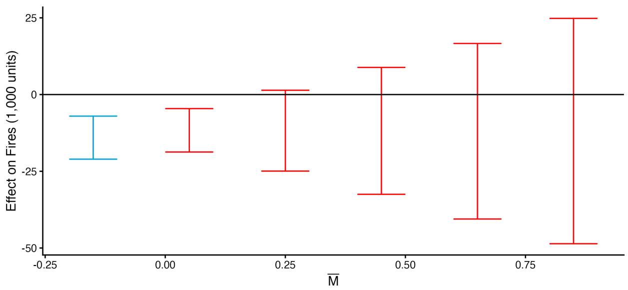
2.1 No moderator



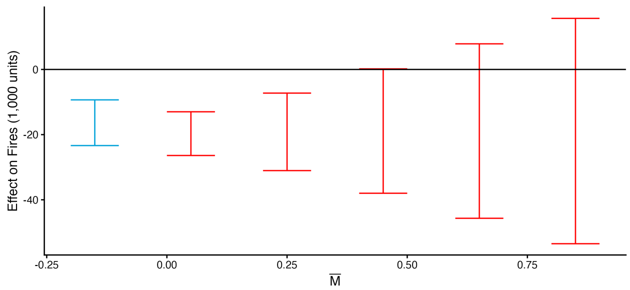
downup_ac — original



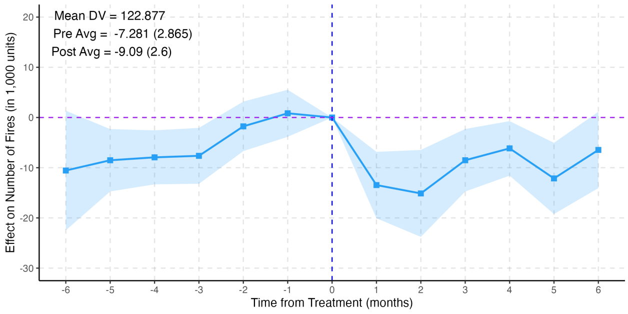
downup_ac — rotated



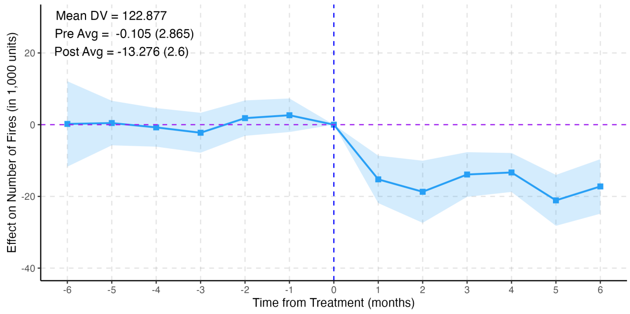
downup_ac — HonestDiD† (original)



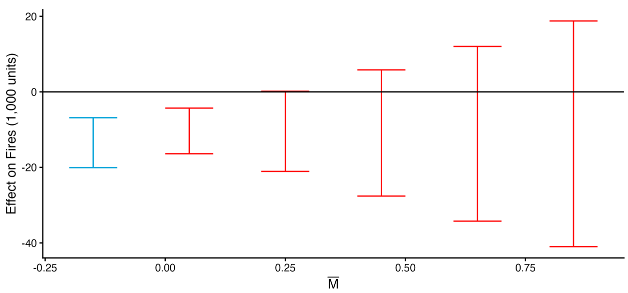
downup_ac — HonestDiD† (rotated)



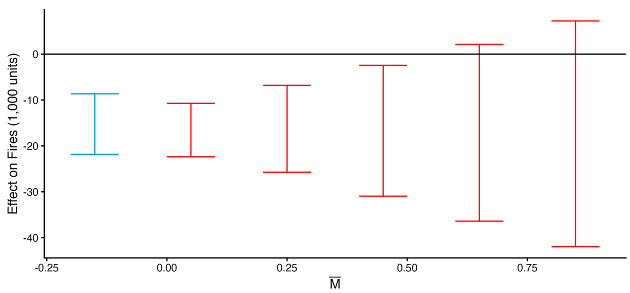
downup_ac_pop — original



downup_ac_pop — rotated



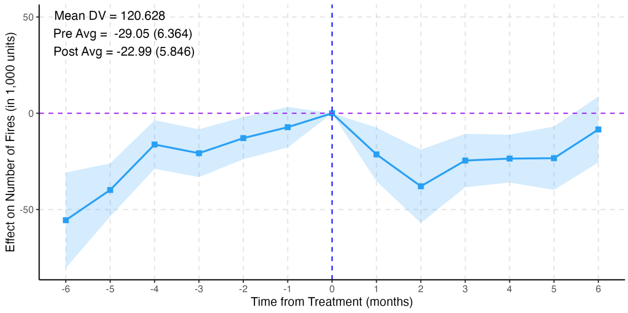
downup_ac_pop — HonestDiD† (original)



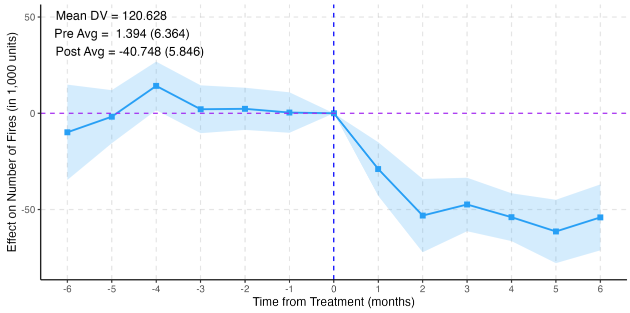
downup_ac_pop — HonestDiD† (rotated)

Figure 1: Main event study (_main_2), no moderator.

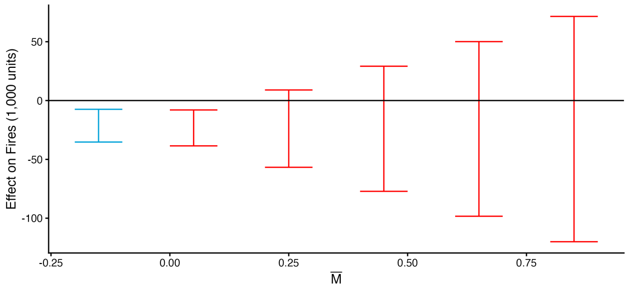
2.2 Rice production moderator (`rice_prod_aclvl_ahigh`)



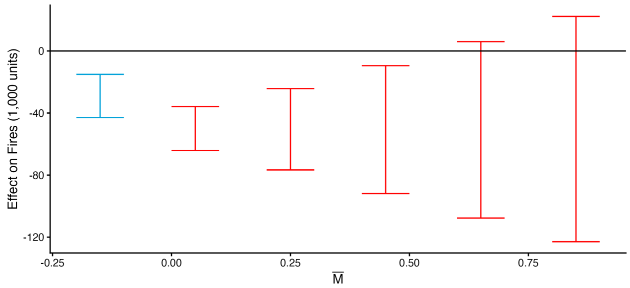
downup_ac — original



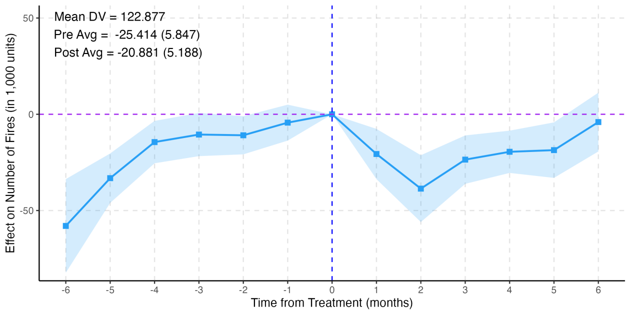
downup_ac — rotated



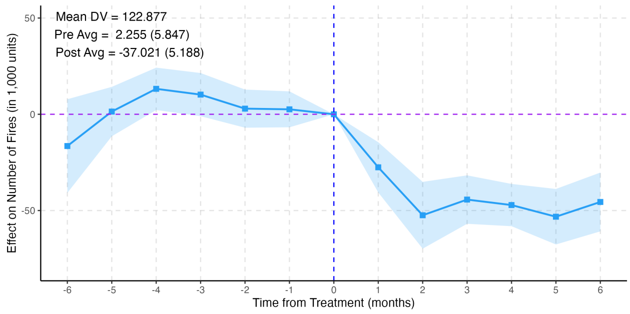
downup_ac — HonestDiD† (original)



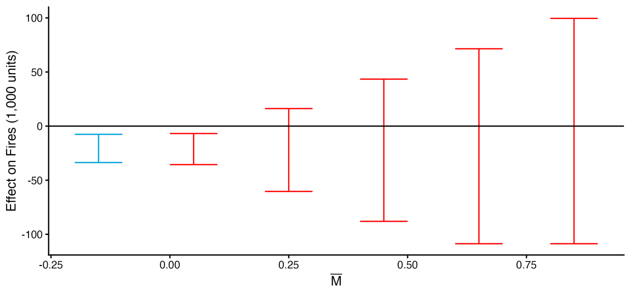
downup_ac — HonestDiD† (rotated)



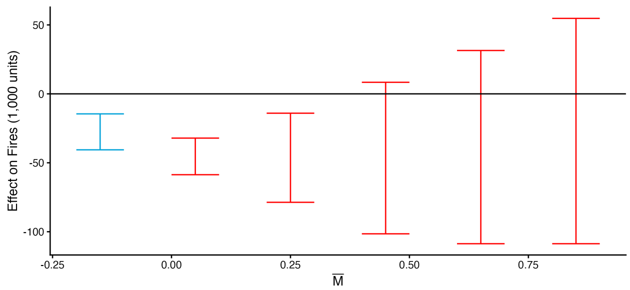
downup_ac_pop — original



downup_ac_pop — rotated



downup_ac_pop — HonestDiD† (original)



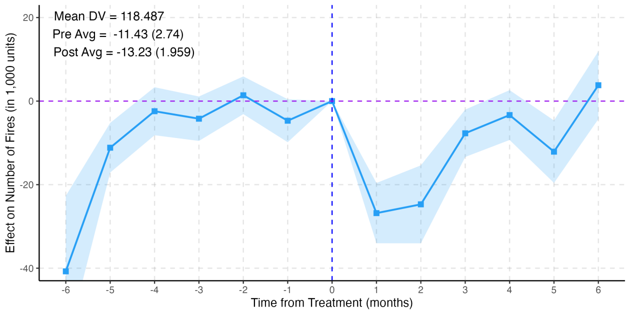
downup_ac_pop — HonestDiD† (rotated)

Figure 2: Main event study (`_main_2`), rice production moderator.

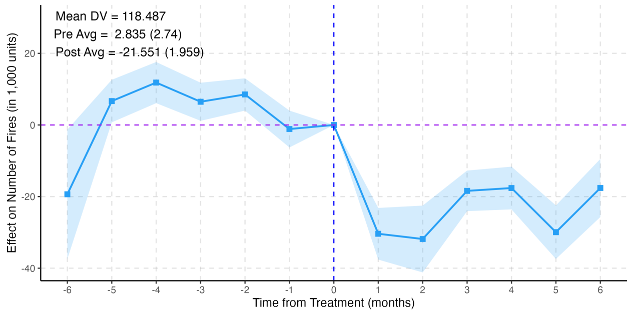
3 Stacked Event Study

Source: `_replication_rural/plotting_event_studies.R` (produces both `downup_ac` and `downup_ac_pop` stacked results). Each row is a 2×2 grid: original | rotated / HonestDiD-ori† | HonestDiD-rotated†.

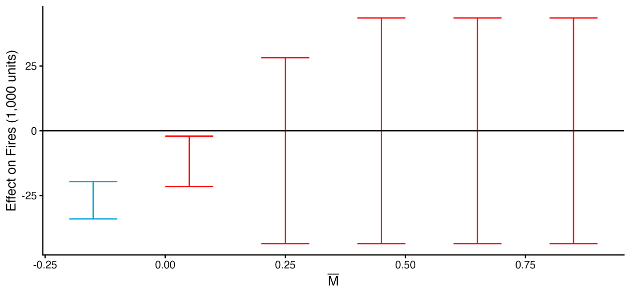
3.1 Unbalanced panel — no moderator



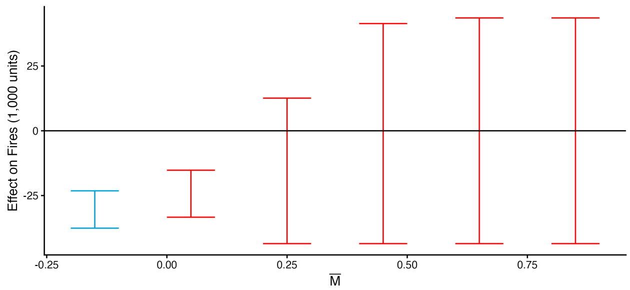
downup_ac — original



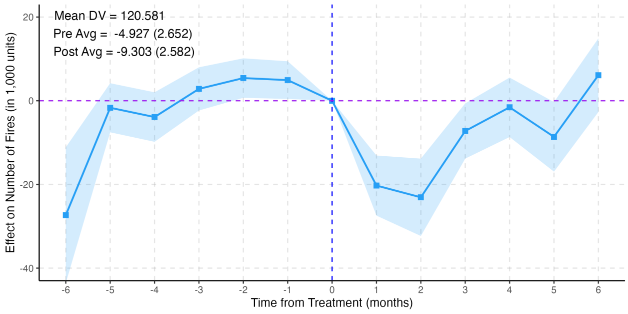
downup_ac — rotated



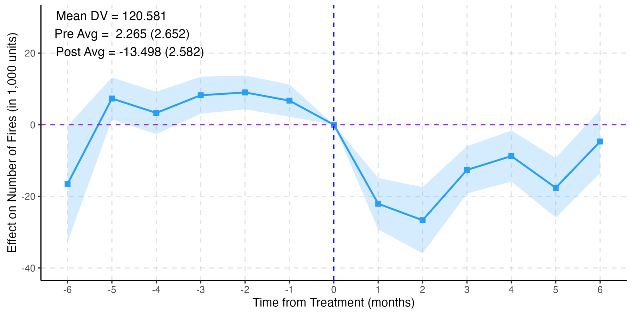
downup_ac — HonestDiD† (original)



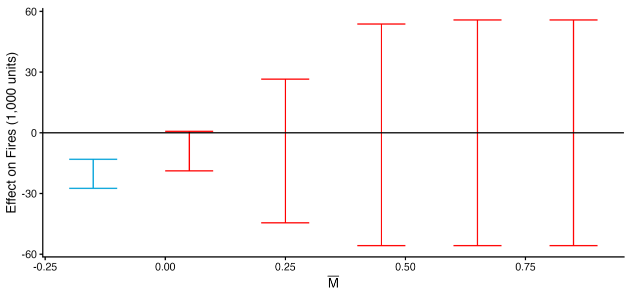
downup_ac — HonestDiD† (rotated)



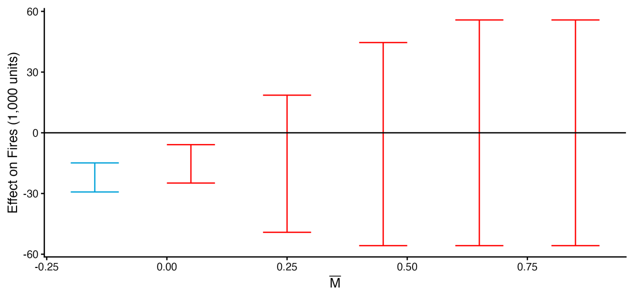
downup_ac_pop — original



downup_ac_pop — rotated



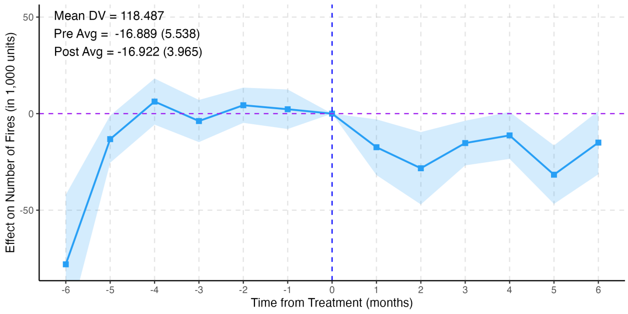
downup_ac_pop — HonestDiD† (original)



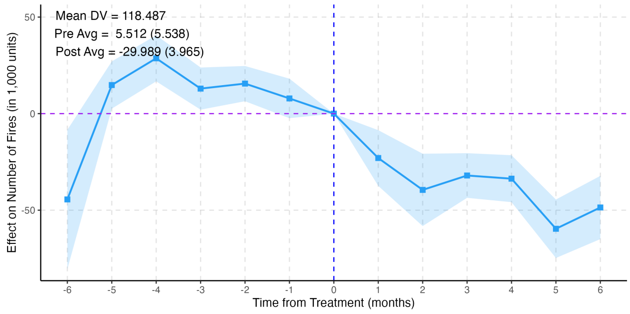
downup_ac_pop — HonestDiD† (rotated)

Figure 3: Stacked event study, unbalanced panel, no moderator.

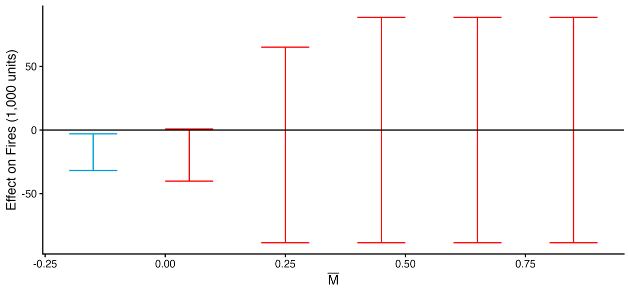
3.2 Unbalanced panel — rice production moderator



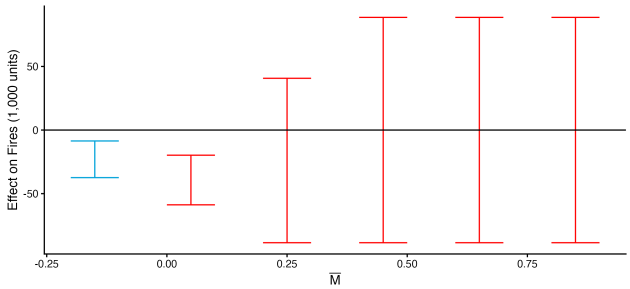
downup_ac — original



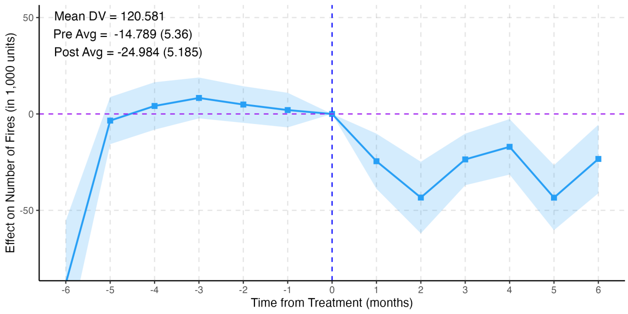
downup_ac — rotated



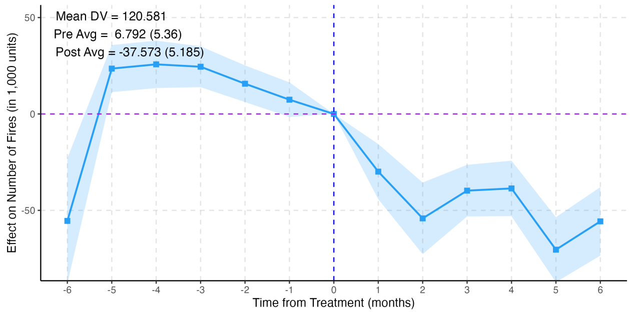
downup_ac — HonestDiD† (original)



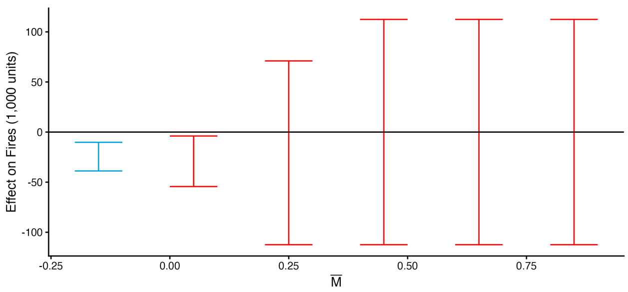
downup_ac — HonestDiD† (rotated)



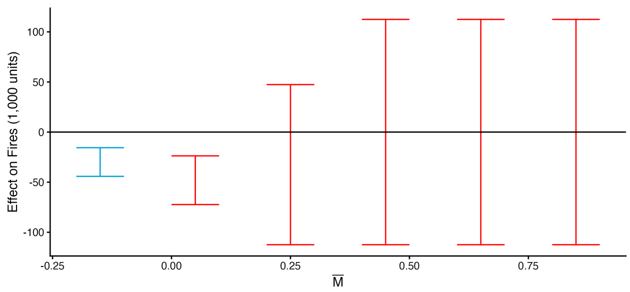
downup_ac_pop — original



downup_ac_pop — rotated



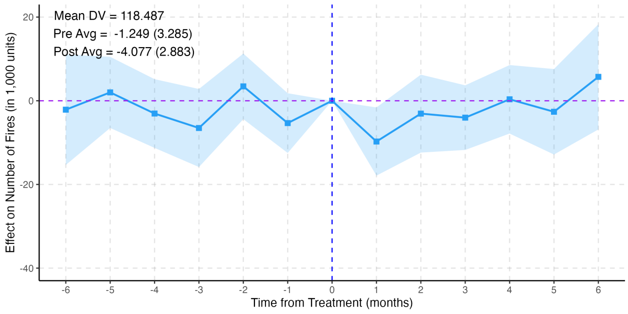
downup_ac_pop — HonestDiD† (original)



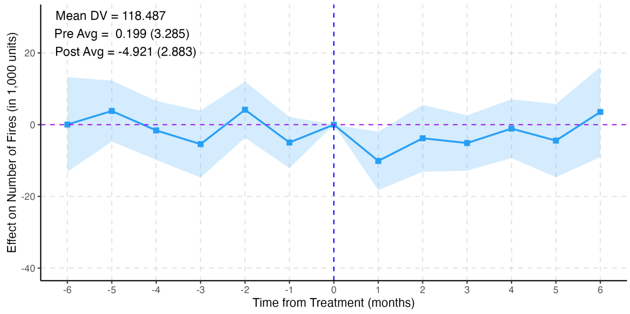
downup_ac_pop — HonestDiD† (rotated)

Figure 4: Stacked event study, unbalanced panel, rice production moderator.

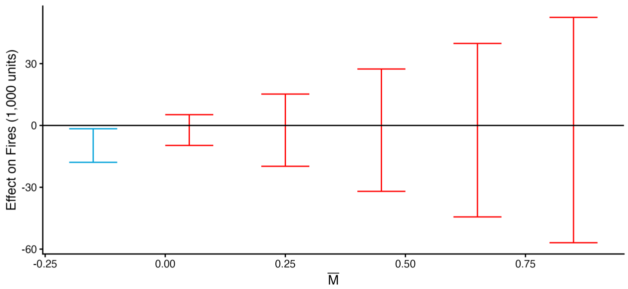
3.3 Balanced panel — no moderator



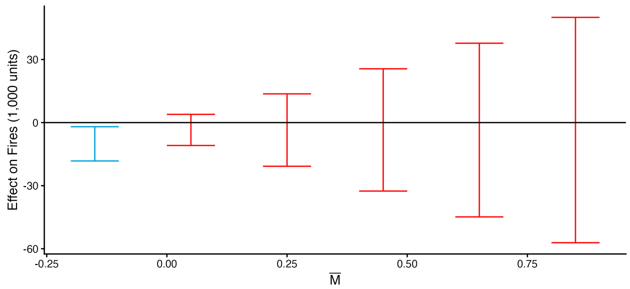
downup_ac — original



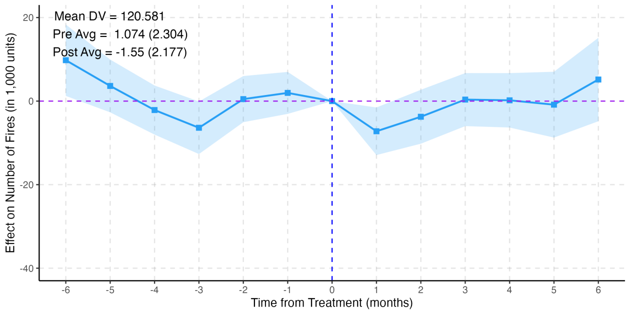
downup_ac — rotated



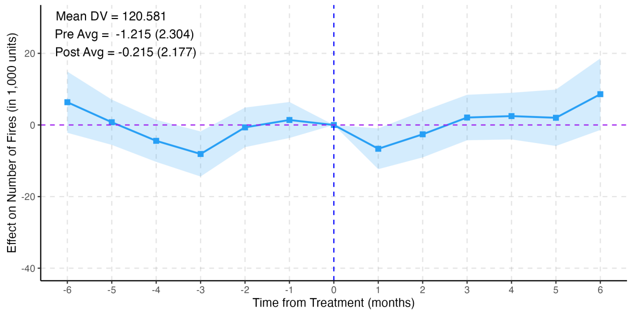
downup_ac — HonestDiD† (original)



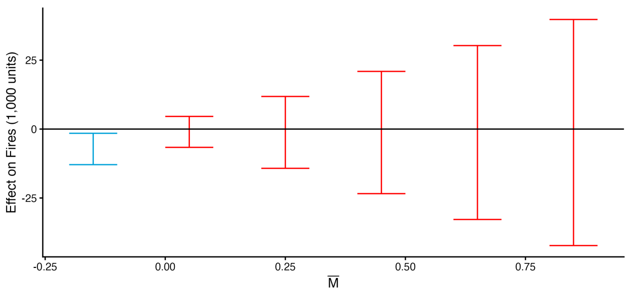
downup_ac — HonestDiD† (rotated)



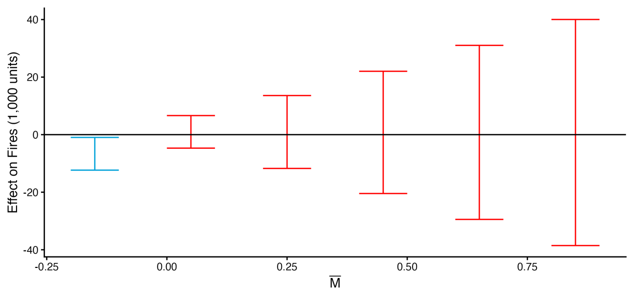
downup_ac_pop — original



downup_ac_pop — rotated



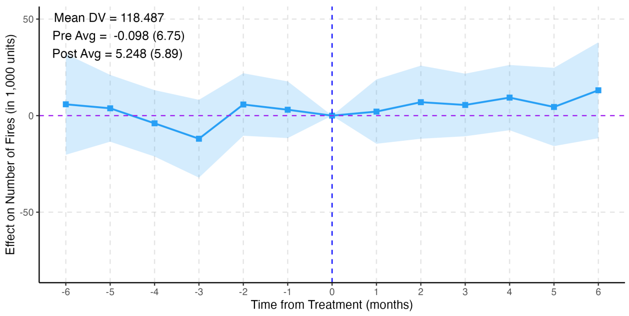
downup_ac_pop — HonestDiD† (original)



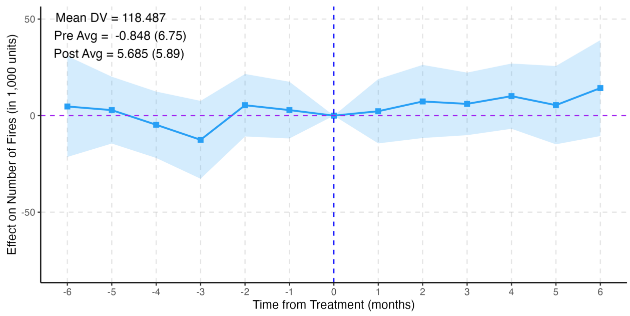
downup_ac_pop — HonestDiD† (rotated)

Figure 5: Stacked event study, balanced panel, no moderator.

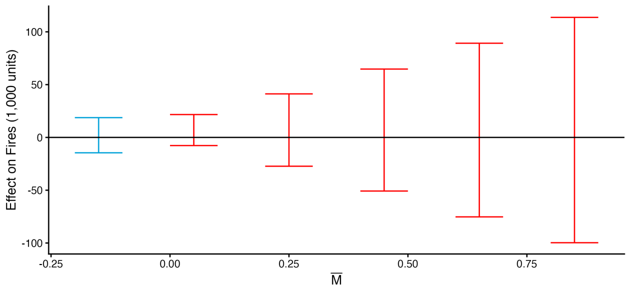
3.4 Balanced panel — rice production moderator



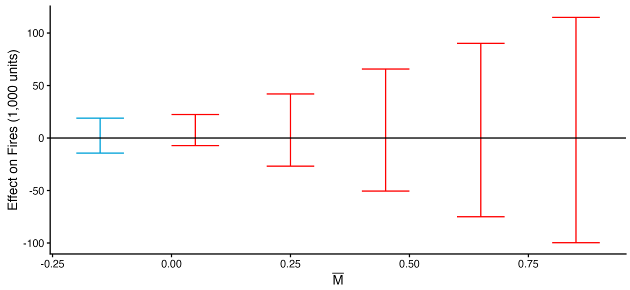
downup_ac — original



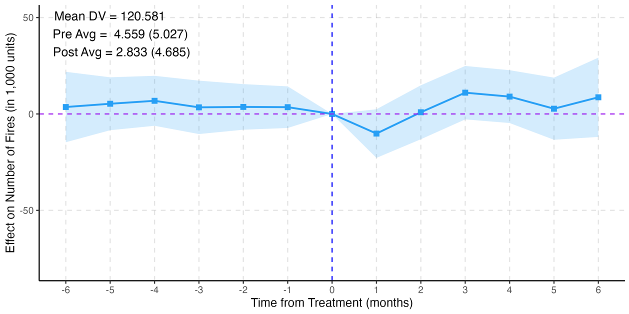
downup_ac — rotated



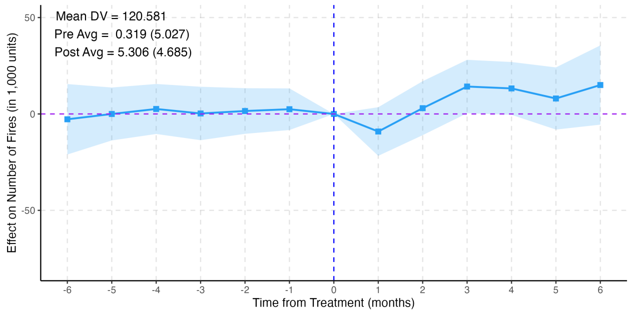
downup_ac — HonestDiD† (original)



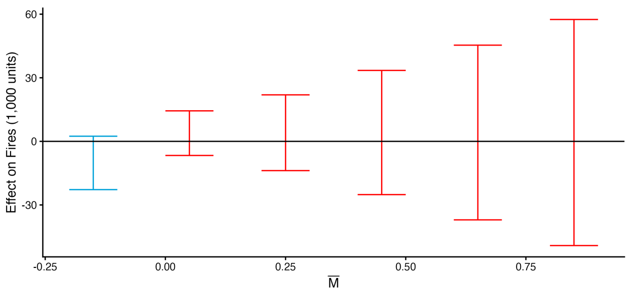
downup_ac — HonestDiD† (rotated)



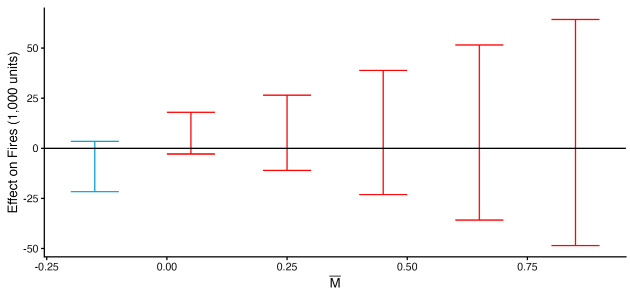
downup_ac_pop — original



downup_ac_pop — rotated



downup_ac_pop — HonestDiD† (original)



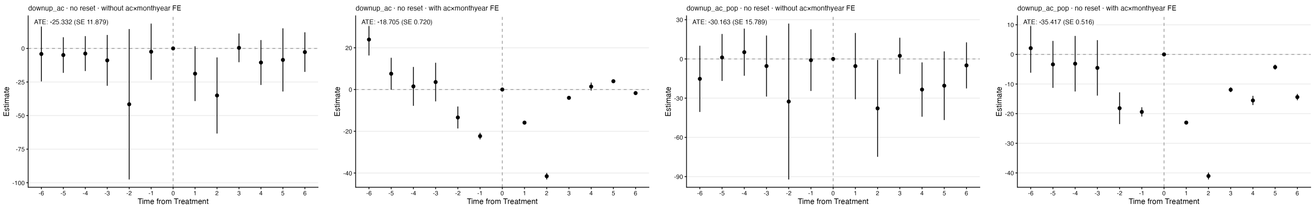
downup_ac_pop — HonestDiD† (rotated)

Figure 6: Stacked event study, balanced panel, rice production moderator.

4 dCDH Event Study (de Chaisemartin & d'Haultfœuille)

Source: `plot_dCDH_event_study.R` (reads per-fit CSVs from `data_output/intermediate/dCDH/`, writes PNGs to `tex/paper/figures/`). Within each panel the top-left annotation reports ATE and SE. Columns within each row: `downup_ac` no ac×monthyear FE | `downup_ac` with ac×monthyear FE | `downup_ac_pop` no FE | `downup_ac_pop` with FE.

4.1 No reset



`downup_ac`, no ac×monthyear FE

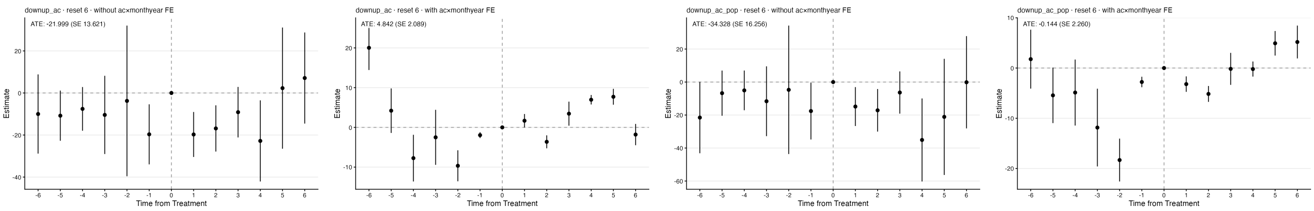
`downup_ac`, with ac×monthyear FE

`downup_ac_pop`, no ac×monthyear FE

`downup_ac_pop`, with ac×monthyear FE

Figure 7: dCDH event study, raw panel (no reset). Effects 1:6, placebos 1:6.

4.2 Reset every 6 same-treatment periods (`apply_reset(6)`)



`downup_ac`, no ac×monthyear FE

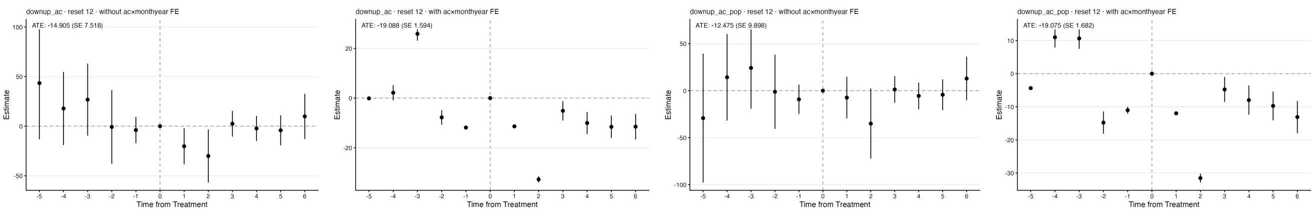
`downup_ac`, with ac×monthyear FE

`downup_ac_pop`, no ac×monthyear FE

`downup_ac_pop`, with ac×monthyear FE

Figure 8: dCDH event study, reset every 6 same-treatment periods. Effects 1:6, placebos 1:6.

4.3 12-month calendar buckets (`generate_12month_subgroups`)



`downup_ac`, no ac×monthyear FE

`downup_ac`, with ac×monthyear FE

`downup_ac_pop`, no ac×monthyear FE

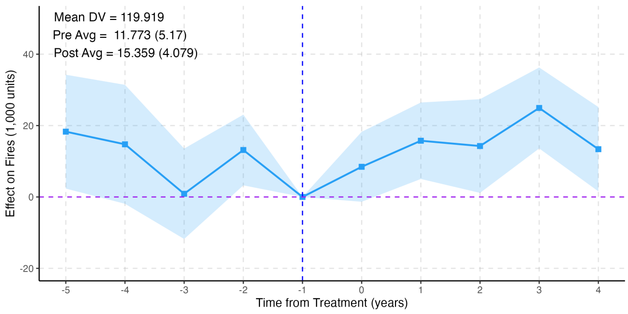
`downup_ac_pop`, with ac×monthyear FE

Figure 9: dCDH event study, 12-month calendar buckets. Effects 1:6, placebos 1:5.

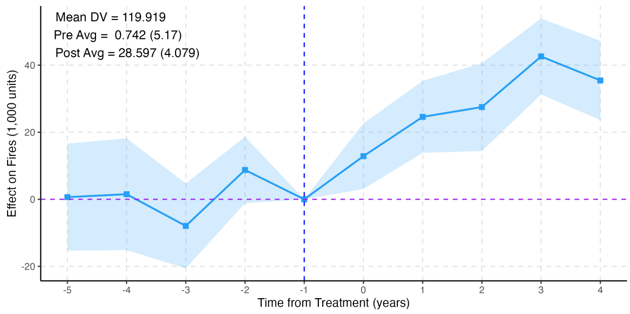
5 App 16: Politician Characteristics Event Study

Source: _replication_rural/plotting_event_studies.R and _replication_rural_acpop/plotting_event_stud
Each row is a 2×2 grid: original | rotated / HonestDiD-ori† | HonestDiD-rotated†.

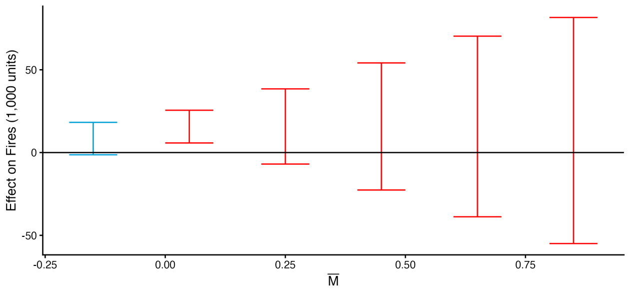
5.1 No moderator



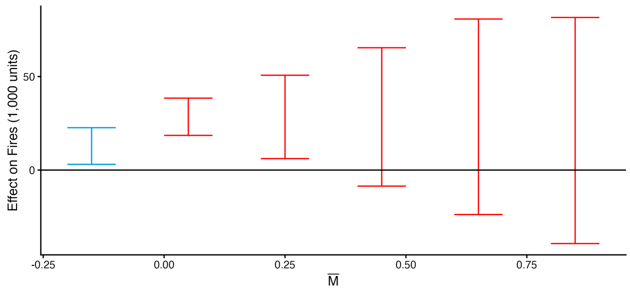
downup_ac — original



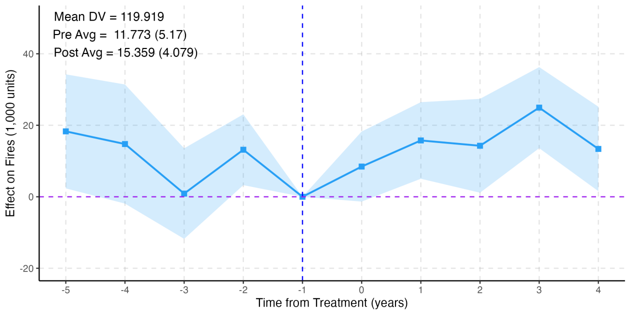
downup_ac — rotated



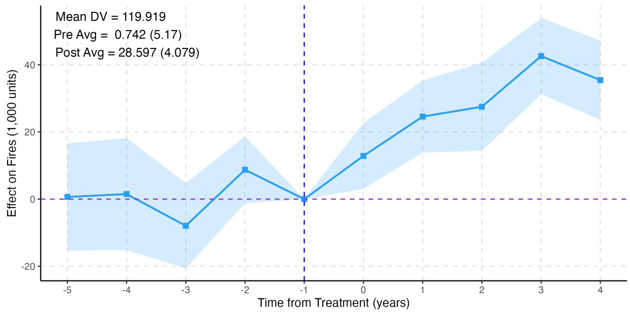
downup_ac — HonestDiD† (original)



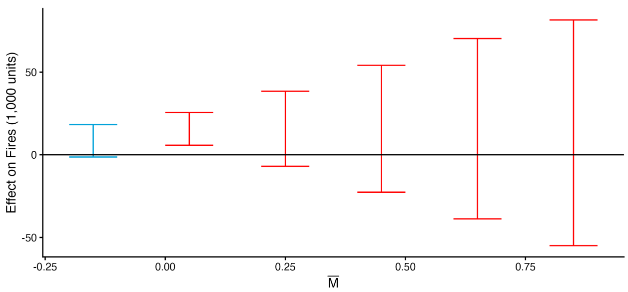
downup_ac — HonestDiD† (rotated)



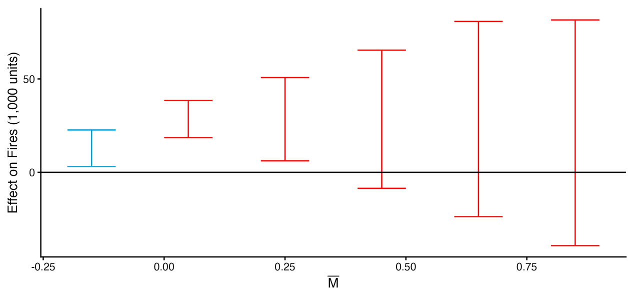
downup_ac_pop — original



downup_ac_pop — rotated



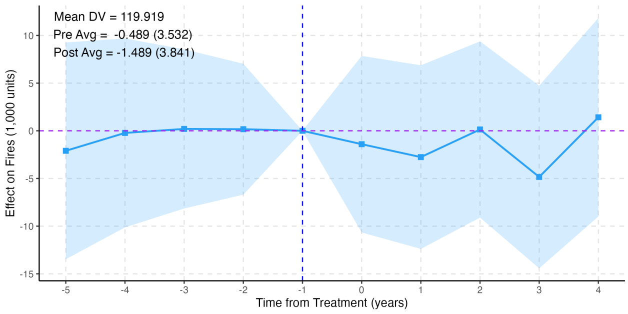
downup_ac_pop — HonestDiD† (original)



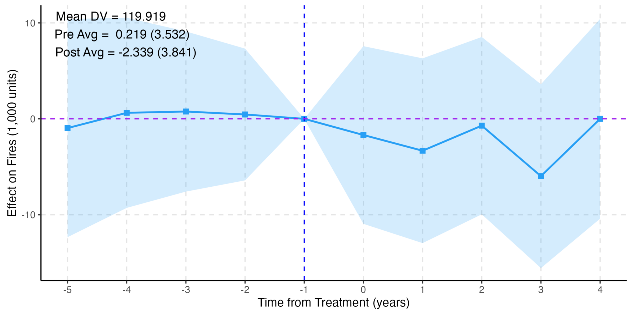
downup_ac_pop — HonestDiD† (rotated)

Figure 10: App 16: politician char. event study, no moderator.

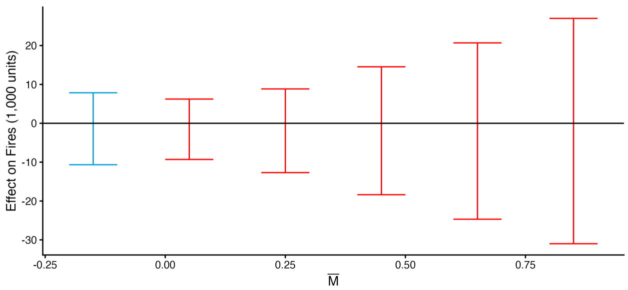
5.2 `downup_ac` moderator



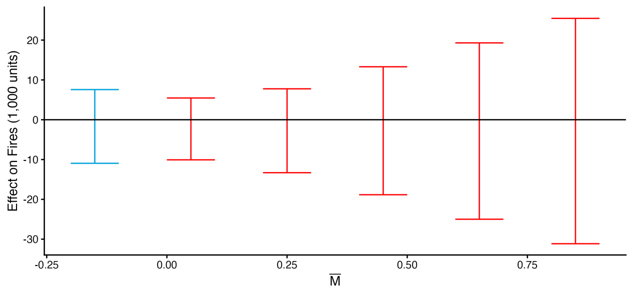
downup_ac — original



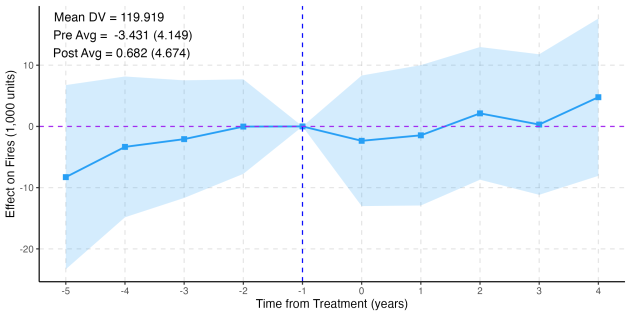
downup_ac — rotated



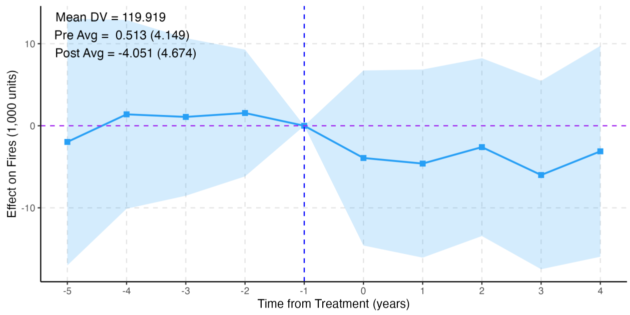
downup_ac — HonestDiD† (original)



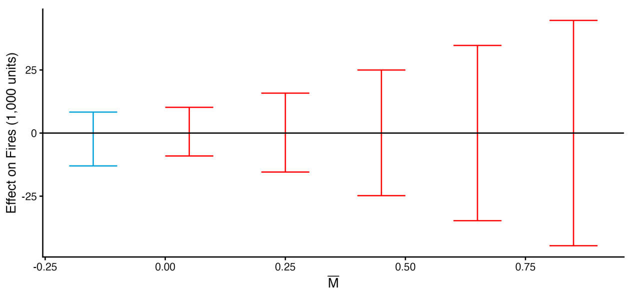
downup_ac — HonestDiD† (rotated)



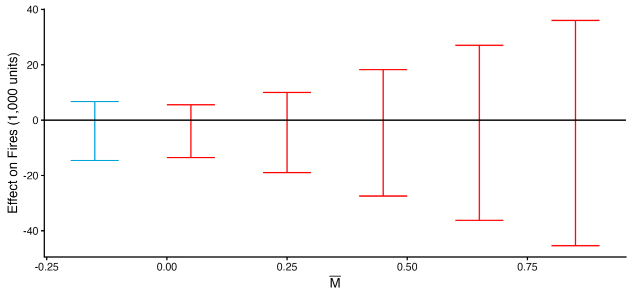
downup_ac_pop — original



downup_ac_pop — rotated



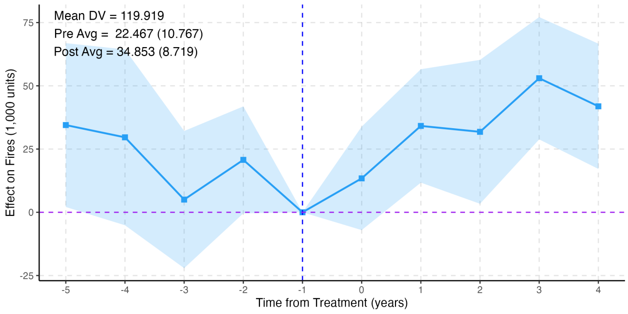
downup_ac_pop — HonestDiD† (original)



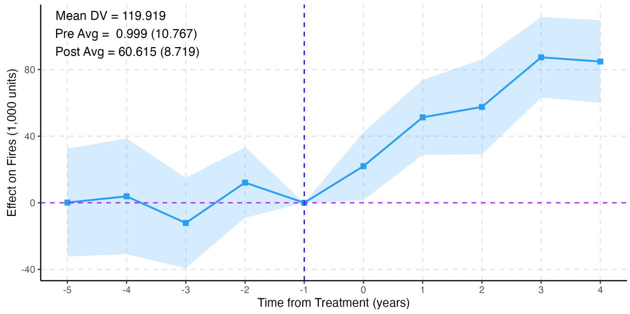
downup_ac_pop — HonestDiD† (rotated)

Figure 11: App 16: politician char. event study, `downup_ac` moderator.

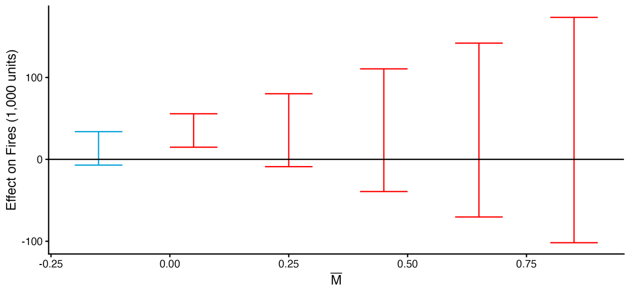
5.3 Rice production moderator



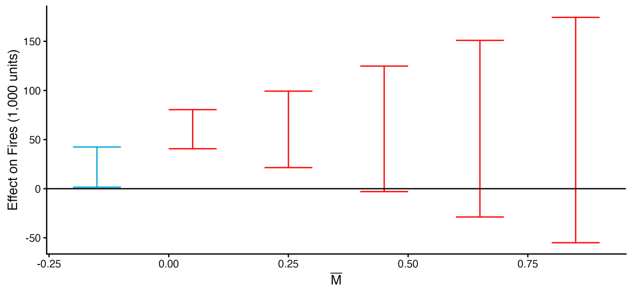
downup_ac — original



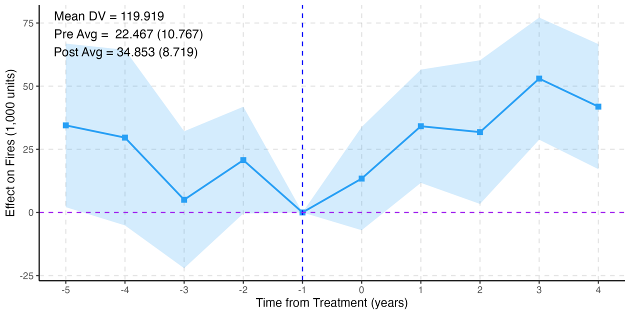
downup_ac — rotated



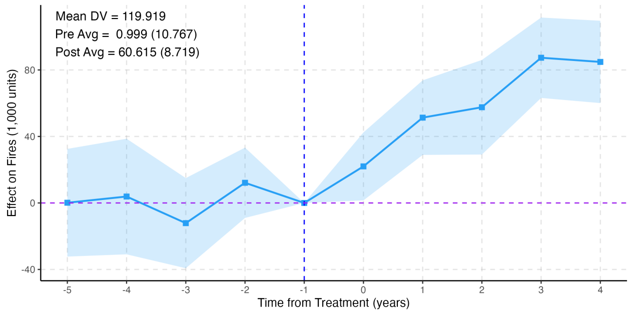
downup_ac — HonestDiD† (original)



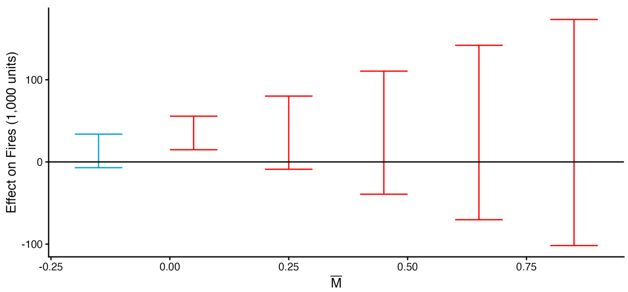
downup_ac — HonestDiD† (rotated)



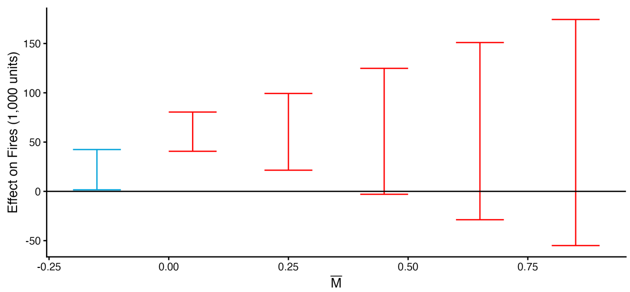
downup_ac_pop — original



downup_ac_pop — rotated



downup_ac_pop — HonestDiD† (original)



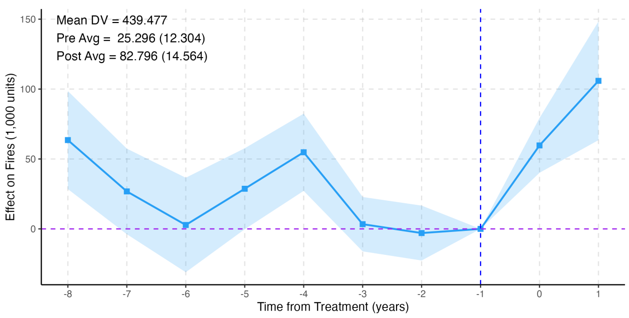
downup_ac_pop — HonestDiD† (rotated)

Figure 12: App 16: politician char. event study, rice production moderator.

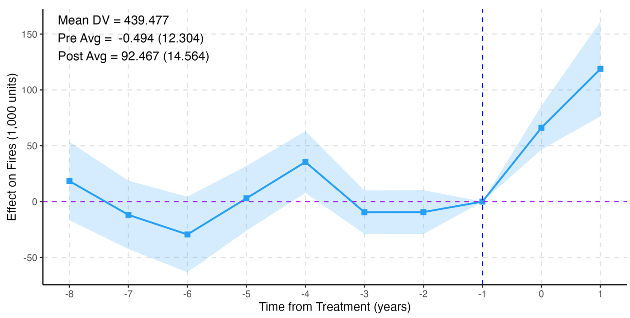
6 App 17: 5 km Protest Event Study

Source: `_replication_rural/plotting_event_studies.R` and `_replication_rural_acpop/plotting_event_stud`
Each row is a 2×2 grid: original | rotated / HonestDiD-ori† | HonestDiD-rotated†.

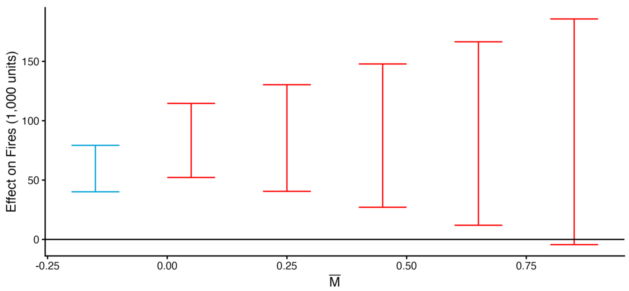
6.1 No moderator



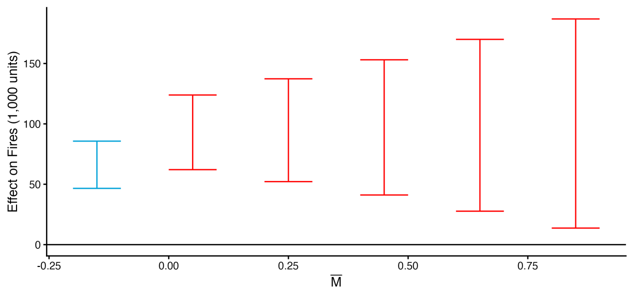
downup_ac — original



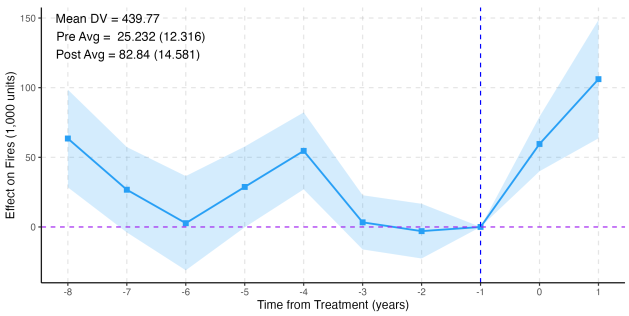
downup_ac — rotated



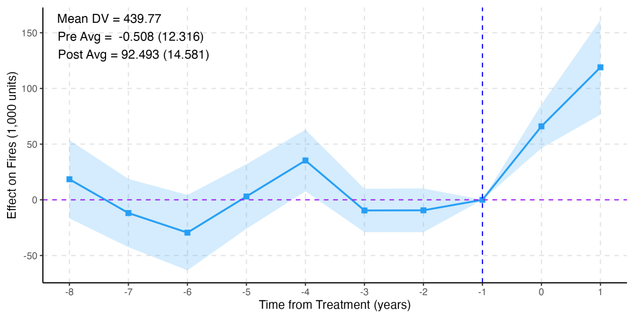
downup_ac — HonestDiD† (original)



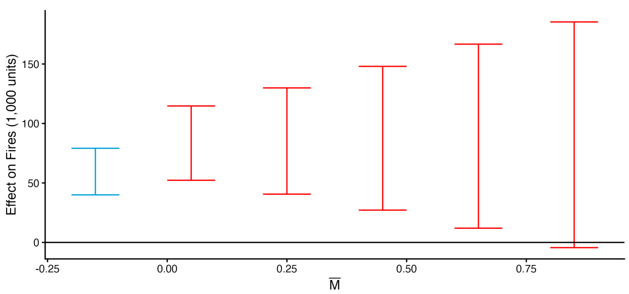
downup_ac — HonestDiD† (rotated)



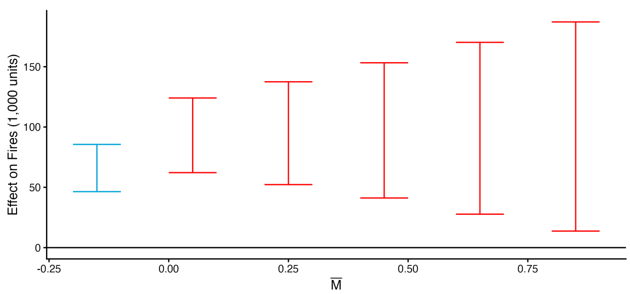
downup_ac_pop — original



downup_ac_pop — rotated



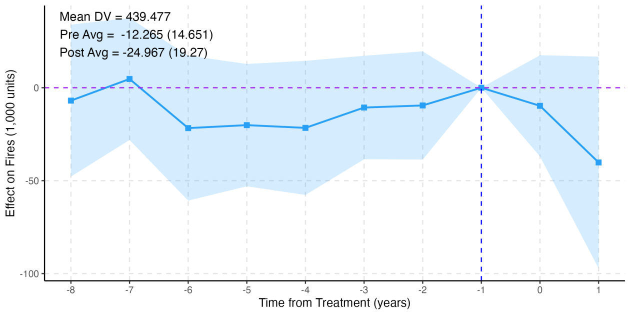
downup_ac_pop — HonestDiD† (original)



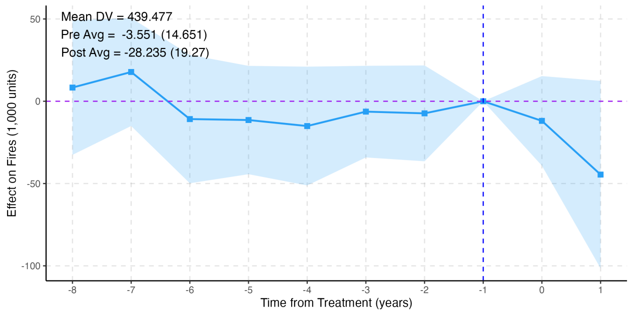
downup_ac_pop — HonestDiD† (rotated)

Figure 13: App 17: 5 km protest event study, no moderator.

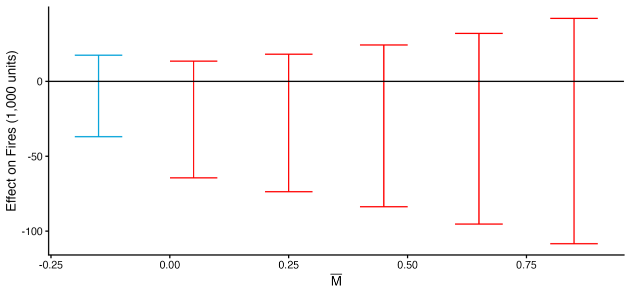
6.2 downup_ac moderator



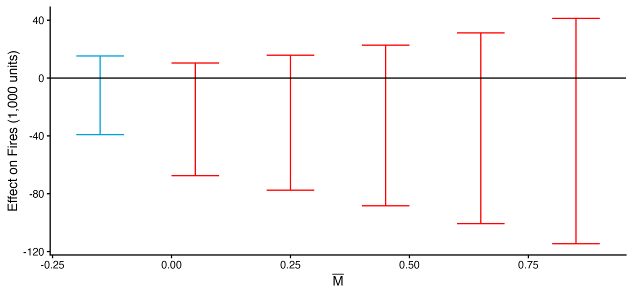
downup_ac — original



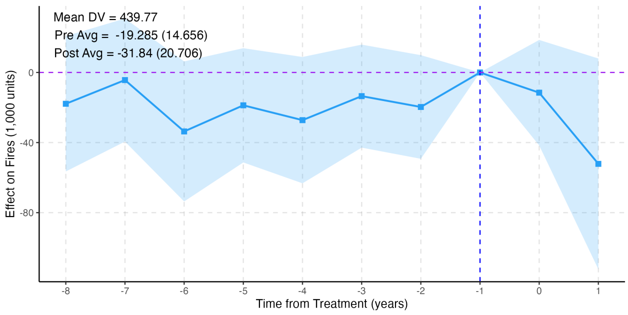
downup_ac — rotated



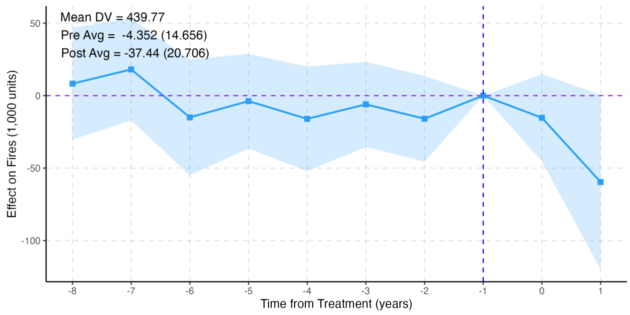
downup_ac — HonestDiD† (original)



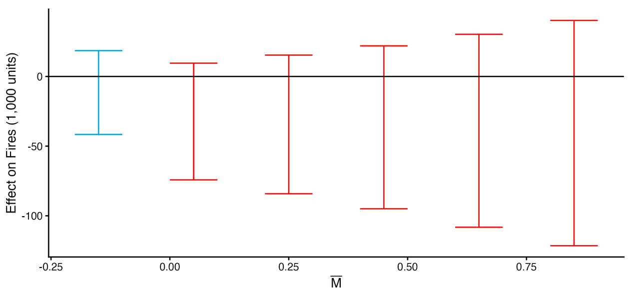
downup_ac — HonestDiD† (rotated)



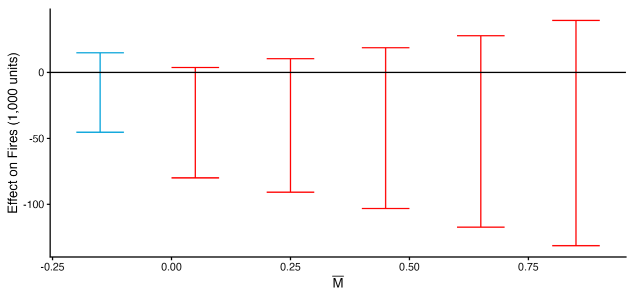
downup_ac_pop — original



downup_ac_pop — rotated



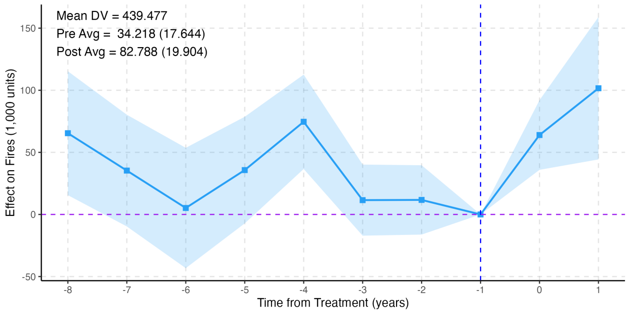
downup_ac_pop — HonestDiD† (original)



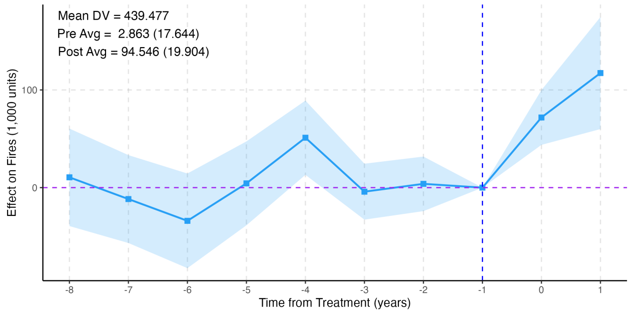
downup_ac_pop — HonestDiD† (rotated)

Figure 14: App 17: 5 km protest event study, downup_ac moderator.

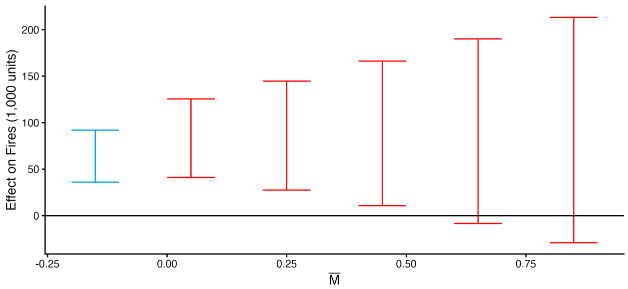
6.3 Rice production moderator



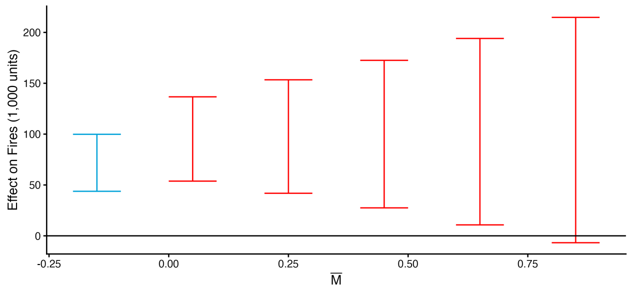
downup_ac — original



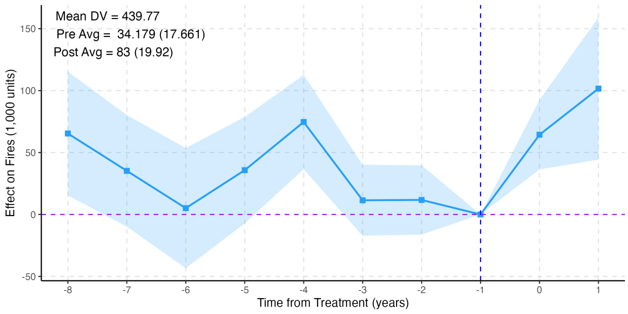
downup_ac — rotated



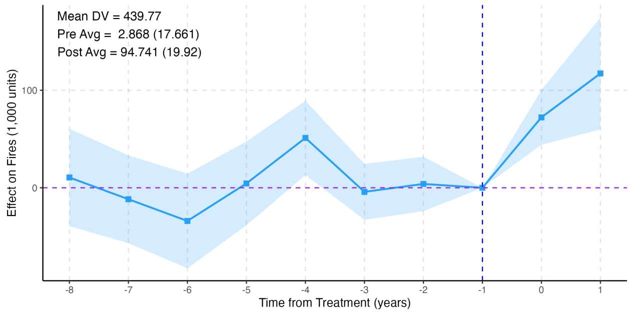
downup_ac — HonestDiD† (original)



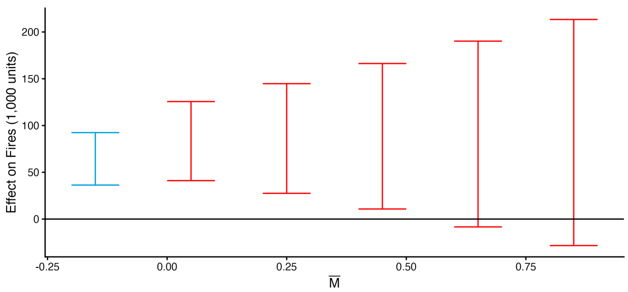
downup_ac — HonestDiD† (rotated)



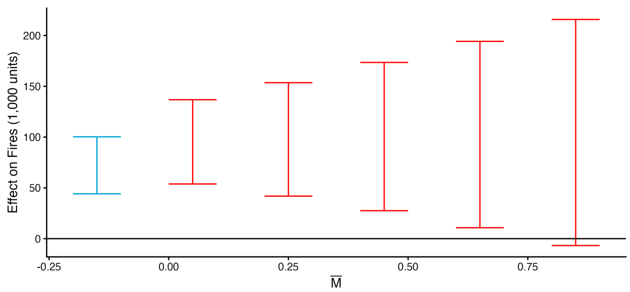
downup_ac_pop — original



downup_ac_pop — rotated



downup_ac_pop — HonestDiD† (original)

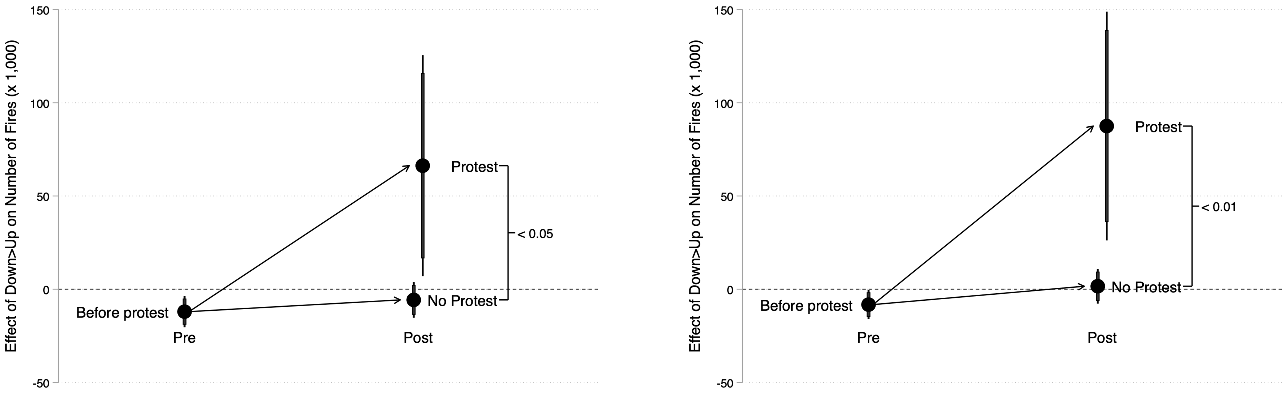


downup_ac_pop — HonestDiD† (rotated)

Figure 15: App 17: 5 km protest event study, rice production moderator.

7 App 18: Protest Downup Interaction Plot

Source: `plotting_interaction.do` via `interaction_graph.ado`. These are pre/post interaction figures (not event-study paths) — there is no ori/rotated/HonestDiD decomposition. The moderating variable is `downup_ac` (left) or `downup_ac_pop` (right).



`downup_ac` `downup_ac_pop`

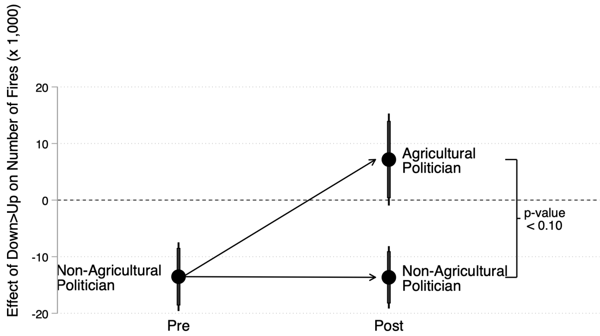
Figure 16: App 18: protest pre/post interaction, `downup` as moderating variable.

8 App 19: Polischar Downup Interaction Plot

Source: `plotting_interaction.do` via `interaction_graph.ado`. The moderating variable is `downup_ac` (left) or `downup_ac_pop` (right).

(pending: $_{a}pp_{1}9_{p}olischar_{f}e12_{d}id_{d}ownup_{i}nter_{p}lot_{r}ural_{1}.png$)

`downup_ac`



`downup_ac_pop`

Figure 17: App 19: polischar pre/post interaction, `downup` as moderating variable.